

PROJECT BRIEF

Web Development 2 PROG322-EC20

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Project Title: Delivery App System using Laravel.
Final Project, 60% of the final grade.

1. Introduction

The Delivery App System is a robust and scalable web-based application designed to connect clients with delivery drivers efficiently. The system enables clients to register, create delivery requests, select drivers manually or allow automatic allocation, track their orders in real time, and make payments via multiple channels, including credit cards and cryptocurrency. On the other hand, drivers can register, provide vehicle details, set their pricing structure, manage assigned deliveries, and update their task status. The admin panel allows administrators to oversee drivers, handle approvals, manage payments, and implement a loyalty points system to reward drivers based on completed kilometers. With real-time notifications, live chat, GPS tracking, and order status updates, this system provides a seamless and efficient delivery service experience for all stakeholders.

2. Aim

This app is designed to streamline and optimize the delivery process by providing an efficient and user-friendly platform that connects clients and drivers while ensuring seamless order management, tracking, and payment processing.

3. Objectives

a) User Roles & Authentication

- **Client Registration**

Clients can register using their email, phone number, or **social media accounts** (such as Google and Facebook).

- **Driver Registration**

Drivers must register manually and provide additional details, such as their vehicle type, plate number, and pricing model.

- **Email & OTP Verification**

To enhance security, users must verify their accounts via email before accessing the platform.

b) Driver-Specific Features

- **Profile Setup (Name, Vehicle Type, Plate Number, etc.)**

Each driver must complete a detailed profile, including their name, contact information, vehicle type, license details, and pricing model.

- **Define Pricing (Fixed price or per km)**

Drivers can set their pricing model as either a fixed rate per delivery or a per-kilometer charge. The system automatically calculates the delivery cost based on package dimensions and weight, ensuring fair pricing for both clients and drivers.

- **Task Management (View and update assigned deliveries)**

A driver can access a personalized dashboard to view their assigned deliveries, check delivery details, and update the status of each task (e.g., Accepted, In Progress, Delivered, or Canceled).

- **Set Availability (Specify working hours and locations)**

Drivers can set their working hours and specify which cities or governorates they serve. This allows clients to find available drivers in their area efficiently.

- **Push Notifications for New Orders**

Whenever a driver is assigned a new delivery request, they receive an instant push notification via Firebase Cloud Messaging (FCM) to ensure quick response times.

- **Earnings & Commission Overview**

Drivers can track their earnings and commissions through a financial dashboard, which provides insights into completed deliveries, total revenue, and pending payments.

- **Specify Delivery Time & Date (Sync to Calendar)**

Drivers can schedule deliveries based on availability, and both the client and driver will have the task automatically added to their calendar for better time management.

c) Client-Specific Features

- **Browse Drivers' Profiles (View ratings & reviews)**

Clients can browse available drivers, check their profiles, read previous customer reviews, and select a driver based on reputation and availability.

- **Create Delivery Requests (Specify package size, weight, etc.)**

Clients can create new delivery requests by specifying the pickup and drop-off locations, package dimensions, weight, and delivery urgency.

- **Choose Driver or Auto-Allocate (Based on area)**
Clients can manually select a driver based on their profile and pricing or opt for an auto-assigned driver based on their location and availability.
- **Order Tracking**
Clients can track their orders status
- **Live Chat with Driver**
Clients can communicate with their assigned driver using a built-in chat system.
- **Multi-Currency Price Display**
The platform supports multiple currencies, allowing clients to view and pay in their preferred currency, depending on their location.
- **Payment Methods:**
 - Credit/Debit Card (Stripe, PayPal, etc.)
 - Cryptocurrency Payments (e.g., Coinbase API, Bitcoin, Ethereum)
 - Cash on Delivery (COD)
- **Delivery Calendar Sync (Client & Driver)**
Once a delivery request is confirmed, it is automatically added to both the client and driver's calendar to prevent scheduling conflicts.
- **Rate & Review Drivers**
After each delivery, clients can rate and review the driver's performance, helping to maintain service quality and accountability.

d) Admin Panel

- **Approve & Manage Drivers**
Admins can approve, suspend, or block driver accounts
- **Oversee Delivery Requests & Orders**
Admins have a dashboard to track all active, completed, or pending orders.
- **Set Loyalty Program Rules (Points per km)**
Admins can configure loyalty rewards for clients
- **Generate Reports on Revenue & Performance**
Admins can generate reports on total earnings, driver performance, client spending, and service demand trends.

e) Bonus

The system provides real-time tracking, allowing clients to follow their deliveries on a live map with estimated time of arrival.

4. Requirements and constraints

- The system should be scalable and modular, allowing for easy integration of additional devices and sensors.
- The design should prioritize user-friendliness, ensuring that homeowners can easily interact with and control the system.
- Technology: Laravel framework, MySQL database, Bootstrap or similar frontend framework.
- Utilization of a ticketing system such as Azure DevOps or Jira for efficient project management, issue tracking, and task assignment.
- The project should be managed using Git version control system, ensuring proper collaboration, version management, and code review processes.
- APIs: Google Maps API for location services, currency exchange API for currency conversion, social media APIs for login integration.
- Security: Implementation of secure authentication and data encryption mechanisms.

5. Deliverables

- Fully functional e-commerce platform developed using Laravel
- Source code repository with version control.
- Presentation and demonstration showcasing the functionality and features of the system.

6. Evaluation Rubric

The project will be assessed based on the following criteria.

Criteria (Based on the CLO/Kpi)	Beginning 1	Progressing 2	Developing 3	Exemplary 4	Coef
Understanding of Concepts	Limited understanding of basic concepts and principles.	Partial understanding of basic concepts and principles.	Demonstrates a solid understanding of basic concepts and principles.	Demonstrates a deep understanding of advanced concepts and principles.	20
Ticketing	Unable to effectively use the ticketing system for issue tracking and project management.	Partially utilizes the ticketing system but lacks organization and clarity in tracking tasks and issues.	Uses the ticketing system for basic issue tracking and task management but may require improvement in organization and communication.	Effectively utilizes the ticketing system for comprehensive issue tracking, task assignment, and communication among team members.	10
Front End Design	Unable to formulate a clear and effective design approach for front-end development tasks.	Partially formulates a design approach but lacks coherence or effectiveness in translating concepts into design elements.	Formulates a design approach for front-end development tasks with some effectiveness, demonstrating understanding of design principles.	Formulates a well-defined and effective design approach for front-end development tasks, demonstrating proficiency in design principles and implementation techniques.	15

Presentation Skills	Unable to deliver a coherent presentation, lacking organization and clarity in conveying ideas.	Partially presents ideas but struggles with maintaining audience engagement and coherence in delivery.	Presents ideas with some effectiveness, demonstrating organization and clarity in communication, but may lack engagement.	Delivers a well-structured and engaging presentation, effectively conveying ideas with clarity, coherence, and audience engagement.	15
Overall Functionality	Demonstrates inadequate functionality, with numerous bugs, errors, or missing features affecting usability.	Partially achieves functionality goals but may have significant issues or limitations that impact overall usability and user experience.	Achieves functionality goals with some effectiveness, providing a satisfactory level of usability and user experience, but may still have minor issues or room for improvement.	Achieves overall functionality goals effectively, delivering a high level of usability and user experience with minimal issues or limitations.	40

7. Project Timeline and Follow-up Schedule

- Project Kickoff: 21/03/2024
- Presentation and Evaluation: last 2 sessions

8. Resources and Support

- Access to well-equipped computer labs and software tools for development and testing purposes, ensuring that team members have the necessary resources to work efficiently.
- Availability of relevant textbooks, research papers, and online resources on e-commerce, Laravel framework, API integration, and other relevant topics to support learning and problem-solving.
- Guidance and support from the project supervisor or instructor, who will be available during designated office hours or via email to provide assistance, answer questions, and offer feedback on project progress.
- Regular progress meetings or checkpoints to review project milestones, address any issues or concerns, and ensure that the project is on track for successful completion.

Please visit the following references:

1. Laravel Framework Documentation
 - Website: <https://laravel.com/>
 - Description: The official documentation for the Laravel framework provides comprehensive guides, tutorials, and references for developers. It covers topics such as installation, routing, authentication, database management, and more.
2. GitHub
 - Website: <https://github.com/>

- Description: GitHub is a web-based platform for version control using Git. It offers features such as repository hosting, collaboration tools, issue tracking, and continuous integration. GitHub serves as a central hub for project management, code review, and collaboration among team members.